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Biodiesel production from waste cooking oil via alkali catalyst and its engine test

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ABSTRACT

Waste cooking oils (WCO), which contain large amounts of free fatty acids produced in restaurants, are collected by the environmental protection agency in the main cities of China and should be disposed in a suitable way. Biodiesel production from WCO was studied in this paper through experimental investigation of reaction conditions such as methanol/oil molar ratio, alkaline catalyst amount, reaction time and reaction temperature which are deemed to have main impact on reaction conversion efficiency. Experiments have been performed to determine the optimum conditions for this transesterification process by orthogonal analysis of parameters in a four-factor and three-level test. The optimum experimental conditions, which were obtained from the orthogonal test, were methanol/oil molar ratio 9:1, with 1.0 wt.% sodium hydroxide, temperature of 50 °C and 90 min. Verified experiments showed methanol/oil molar ratio 6:1 was more suitable in the process, and under that condition WCO conversion efficiency led to 89.8% and the physical and chemical properties of biodiesel sample satisfied the requirement of relevant international standards. After the analysis main characteristics of biodiese sample, the impact of biodiesel/diesel blend fuels on an YC6M220G turbo-charge diesel engine exhaust emissions was evaluated compared with 0# diesel. The testing results show without any modification to diesel engine, under all conditions dynamical performance kept normal, and the B20, B50 blend fuels (include 20%, 50% crude biodiesel respectively) led to unsatisfactory emissions whilst the B′20 blend fuel (include 20% refined biodiesel) reduced significantly particles, HC and CO etc. emissions. For example CO, HC and particles were reduced by 18.6%, 26.7% and 20.58%, respectively.

1. Introduction

With the exception of hydroelectricity and nuclear energy, the majority of the world energy needs are supplied through petrochemical sources, coal and natural gas. All of these sources are finite and at current usage rates will be consumed one day in the near future [1]. The depletion of world petroleum reserves and increased environmental concerns has stimulated recent interest in alternative sources for petroleum-based fuels. Biodiesel [2], derived from vegetable oil or animal fats by transesterification with alcohol like methanol and ethanol, is recommended for use as a substitute for petroleum-based diesel mainly because biodiesel is an oxygenated, renewable, biodegradable and environmentally friendly biofuel with similar flow and combustion properties and low emission profile [3,4]. Because of the good properties and the environment improvement due to it, many countries pay much attention to R&D of biodiesel industry and constitute

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favorable legislation for it. More than 2.7 million tons biodiesel in 2003 was made in Europe, and 8–10 million tons is expected in 2010, accounting for 5.75% among the total diesel market, and 20% among the total diesel market in 2020. The capacity of biodiesel production in USA reaches 221,000 tons in 2002, and 1.15 million tons is expected in 2011 and 3.3 million tons in 2016. As world petroleum prices rise, China becomes increasingly reliant on imported fuels and the government has since stepped in to boost the fledgling biodiesel industry. In 2004, the Ministry of Science launched its biofuel technology development project; the following year, the government initiated a special agricultural and forestry biomass development program, setting a nationwide target for annual biodiesel production of 2 million tons by 2010 and 12 million tons by 2020.

Currently, compared to petroleum-based diesel, the high cost of biodiesel is a major barrier to its commercialization. It is reported that approximately 70%–85% of the total biodiesel production cost arises from the cost of raw material. Use of low-cost feedstock such as WCO should help make biodiesel competitive in price with petroleum diesel. Numerous studies have been conducted on biodiesel production and emission testing in the past two decades. Most of the current challenges are targeted to reduce its production cost, as the cost of biodiesel is still higher than its petro-diesel counterpart. This opens a golden opportunity for the use of WCO as its production feedstock. Everywhere in the world, there is an enormous amount of waste lipids generated from restaurants, food processing industries and fast food shops everyday. In China, with annual consumption of edible oils approaching 22 million tons, the country generates more than 4.5 million tons of used oil and grease each year, roughly half of which could be collected through the establishment of an integrated collection and recycling system. Those 2 million tons of "ditch oil" alone would guarantee the smooth operation of all current biodiesel production lines. Reusing of these waste greases cannot only reduce the burden of the government in disposing the waste, maintaining public sewers and treating the oily wastewater, but also lower the production cost of biodiesel significantly. Furthermore, biodiesel fuel has been demonstrated to be successfully produced from waste edible oils by an alkali-catalyzed transesterification process [5–8], and can be considered as alternative fuels in diesel engines and other utilities [9–12]. Our purpose is to find the most appropriate parameters for WCO transesterification reaction process. Consequently, the emissions and performance of biodiesel from WCO running on diesel engine based on different blend fuels was investigated.

2. Materials and experimental methods

2.1. Materials

The WCO was obtained from Yizhong Western & Eastern Restaurant in Tianjin city. Every day this restaurant produces many WCO which used for frying beefsteak, French chips and cooking various Chinese dishes. So the WCO contain some food particles, phospholipids, grease and wax etc. Some chemical properties and fatty acid compositions of WCO and some pure vegetable oils which obtained from Ref. [13] are summarized in Table 1. Fatty acid compositions of WCO were determined by GC. Identification of fatty acids contained in WCO was performed by comparison of retention times with fatty acids standard purchase from Sigma Chemical Co. Ltd. The viscosity of WCO and primary products from methanolysis all were measured with a viscometer (Model SYD-265D-I, Shanghai Changji Geological Apparatus Co. Ltd, China). The WCO showed viscosities higher than pure vegetables oils. The high viscosity of WCO may have influenced negatively the conversion efficiency since it limits the well mixing of substrates. Methanol, ethanol, sulfuric acid, sodium hydroxide commercial grade as a catalyst, dry magnesium sulphate and other chemicals of analytical grade were purchased from Kewei Co. Ltd of Tianjin University.

2.2. Methanolysis

200 ml WCO sample was heated and filtered under vacuum to remove any different solid impurities. Then 40% phosphate acid washing and distillated water washing were adopted to eliminate phospholipids. And in order to avoid saponification reaction for high free fatty acid (FFA) content, the FFA was esterified with methanol by sulfuric acid. When the FFA content was lower than 1.0%, the sulfuric acid was drained and the sodium hydroxide was introduced into the system to complete the transesterification. The treated oil and an appropriate volume of methanol with sodium hydroxide as a catalyst were placed into a dry reaction flask equipped with reflux condenser and magnetic stirrer. Reaction mixture was blended for 60 min at a temperature of 50 °C. The crude ester layer was separated from the glycerol layer in a separating funnel. The crude ester layer consisted of methyl ester, possibly of unreacted oil, methanol, glycerol, catalyst residue and small amount of produced soap. To separate methanol, the crude ester phase was washed three times with distilled water at 50 °C in a separatory funnel, until the washings were neutral. The ester layer was dried by using anhydrous magnesium sulfate and filtered.

Theoretically, the stoichiometry of the transesterification reaction requires 3 mol of alcohol per mol of triglyceride to yield 3 mol of fatty esters and 1 mol of glycerol (see Fig. 1). Since the reaction is reversible, excess alcohol was used to shift the equilibrium to the products side and result in higher ester

Table 1 – Comparison of chemical properties and fatty acid composition of WCO and some vegetable oils

Property	WCO	Cottonseed oil l	Rapeseed oil	Soybean oil
Fatty acid composition (%)				
Palmitic acid C16:0	16	11.67	3.49	11.75
Stearic acid C18:0	5.21	0.89	0.85	3.15
Oleic acid C18:1	34.28	13.27	64.4	23.26
Linoleic acid C18:2	40.76	57.51	22.3	55.53
Linolenic acid C18:3	0	0	8.23	6.31
Specific gravity	0.925	0.912	0.914	0.92
Viscosity (mm^2/s) at 40 °C	66.6	50	39. 5	65
Acid value (mg KOH/g)	7.25	0.11	1.14	0.2

yield. We investigated the role of substrate molar ratio in methanolysis of WCO in solvent-free medium, conducting the reactions at 3:1, 5:1, 6:1, 7:1 and 8:1 methanol/oil molar ratios at 50 °C and 60 min. The sodium hydroxide content was 1.0 wt.% based on the treated oil weight. In order to determine the effect of sodium hydroxide amount on WCO conversion efficiency, reactions were conducted with 6:1 methanol/oil molar ratio at 50 °C with sodium hydroxide amount at 0.5 wt.%, 0.7 wt.%, 1.0 wt.%, 1.1 wt.% and 1.2 wt.% based on treated oil weight. The reaction time was kept constant at 60 min in all experiments. The effect of reaction time on WCO conversion efficiency, the treated oils was alcoholyzed at 50 °C extending the reaction time to 30 min, 50 min, 60 min, 70 min, 90 and 110 min keeping the sodium hydroxide amount and methanol/oil molar ratio at 1.0 wt.% and 6:1, respectively. Finally, the effects of reaction temperature on WCO conversion efficiency at 30 °C, 40 °C, 45 °C, 50 °C, 60 °C and 70 °C were also discussed.

2.3. Gas-chromatography analysis

The biodiesel sample was taken to analyze the purity by gas chromatography (Agilent 6890), equipped with an HP Innowax capillary column (30 m×0.25 mm) and a flame ionization detector (FID). Nitrogen was used as the carrier gas at a constant flow rate of 20 ml/min. The column oven temperature was programmed from 50 to 130 °C (at the rate of 20 °C/min) and held at 130 °C for 5 min, then raised to 260 °C at 2.5 °C/min and maintained at this temperature for 10 min.

2.4. Engine testing

Engine testing was done in Tianjin University State Key Lab for Internal Combustion Engine. A turbocharged YC6M220G heavy duty diesel engine was used. Its main characteristics are shown in Table 2. Relevant engine operation and performance parameters, as well as emissions were recorded. All runs started with a 20-min warm-up period prior to data collection. In accordance with DB 11/185-2003 "Limits and measurement methods for exhaust Pollutant from non-road Diesel Engines" issued by Beijing City Environmental Protection Agency, regulated emission were measured in steady-state 13 mode cycles. The data measured during the tests included coolant and exhaust temperatures, engine speed, brake torque, fuel-to-air ratio, exhaust emissions including carbon monoxide, nitrogen oxide, unburned hydrocarbons, and smoke/soot. PM was sampled by DLS-2300 (Horiba, Japan). Other gases including carbon monoxide (CO), nitrogen oxides (NOx) and unburned hydrocarbons (HC), in addition to the air excess number were measured by a MEXA-7100DEGR (Horiba, Japan). Samples of engine exhaust were drawn for emissions analysis. Because of limited experimental ability, two kinds of biodiesel were used. One was crude biodiesel which neutral by acid then washed by water and may has some methanol and impurities, and the other one was refined biodiesel which was achieved by reduced pressure distillation. Experiments were conducted with different volume proportions of biodiesel/diesel blend fuels, for crude biodiesel, 50:50 (v/v, B50), 20:80 (v/v, B20), and for refined biodiesel, 20:80(v/v, B′20), in addition to 100% 0# diesel (B0), which served as the baseline (reference) fuel.

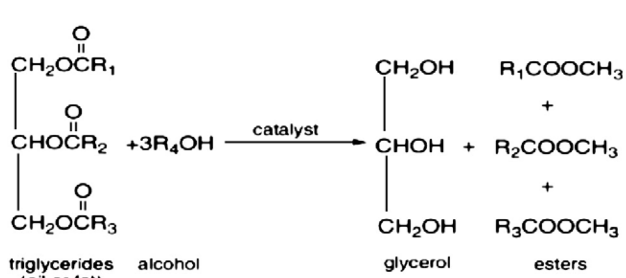

Fig. 1 – Transesterification of triglycerides with alcohol.

Table 2 – Main characteristics of the YC6M220G diesel engine

Specifications	YC6M220G heavy duty diesel engine
Type of engine	6 cylinders, 4-stroke, direct injection
Rated Power	162 kW
Rated speed	2200 r/min
Bore × stroke	120 mm × 145 mm
Displacement	9.839
Volumetric compression ratio	17.5
Valves per cylinders	6
Maximum torque	900 N·m
Maximum brake specific fuel consumption	≤206 g/(kW·h)
Cooling system	water-cooled

3. Results and discussion

3.1. Effect of methanol to oil molar ratio

The molar ratio of methanol to oil is one of the most important variables that affect conversion efficiency as well as production cost of biodiesel and the result is given in Fig. 2. In the experiment, we found the addition of methanol/oil molar ratio <1.5 resulted in the creation of a unique and foamy layer. Molar ratio up to 2.5 produced two layers, although the lower layer was gelatinous and the upper layer was opaque, because of the presence of unreacted triglycerides. Thus, indicating that methanol was insufficient to perform a complete reaction. With the methanol/oil molar ratio increasing, WCO conversion efficiency will be correspondingly increased. The maximum conversion efficiency (88.9%) was achieved at 7:1 methanol/oil molar ratio. With further increase in molar ratio the conversion efficiency more or less remained the same. The WCO conversion efficiency (88.4%) at 6:1 methanol/oil molar ratio was similar to the results obtained by 7:1 methanol/oil molar ratio. Considering that excessive methanol needs to remove from higher methanol/oil molar ratio, so 6:1 methanol/oil molar maybe is more suitable in practical process.

3.2. Effect of NaOH amount

The amount of catalyst used in the process is another variable to take into account, because it not only determines the reaction

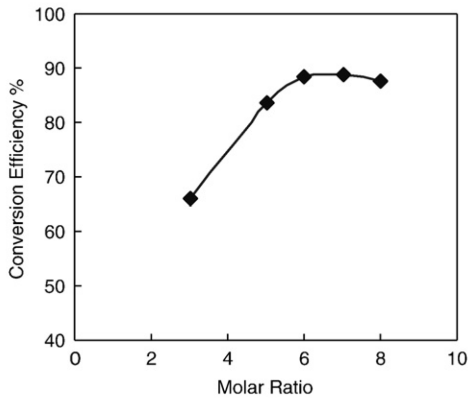

Fig. 2 – Effect of methanol/oil molar ratio on WCO conversion efficiency. Under 1.0 wt.% NaOH, at 50 °C reacted 60 min.

rate, but cause hydrolysis and saponification. Both reactions interfere with the separation of the glycerol rich phase and with the methyl esters purification [14]. The effect of sodium hydroxide amount on WCO conversion efficiency is presented in Fig. 3. From the ranges 0.5–1.0 wt.%, the WCO conversion efficiency increased proportionally with increasing sodium hydroxide amount. The maximum WCO conversion efficiency (85.0%) was observed at 1.0 wt.% sodium hydroxide. Addition of excess amount catalyst, gave rise to the formation of an emulsion, which increased the viscosity and led to the formation of gels. Thus sodium hydroxide amount beyond 1.0 wt.% was not necessary.

3.3. *Effect of reaction time*

Most investigators for alkaline methanolysis have used reaction time of from 1 h to 4 h [15,16], and temperatures at either 60 °C or 65 °C, although some investigators suggested lower temperatures. Effect of reaction time on the WCO conversion efficiency is shown in Fig. 4. To achieve perfect contact between the reagents and the oil during reaction, they must be stirred well at constant rate and well mixed together. The WCO conversion efficiency rapidly increased with the reaction time ranges between 30 min and 60 min, after that, the conversion efficiency kept rising very slowly and then practically constant above 86% at 90 min. So the reaction approved to be in equilibrium and the rate was quite slow.

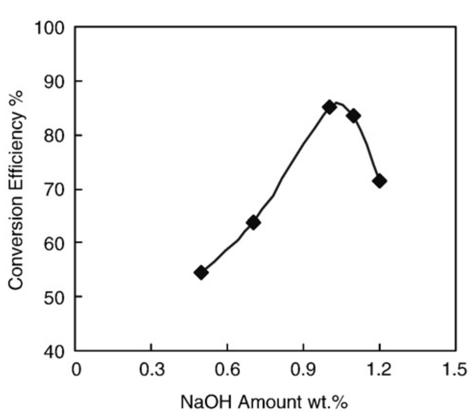

Fig. 3 – Effect of NaOH amount on WCO conversion efficiency. Under 6:1 methanol/oil molar ratio, at 50 °C, reacted 60 min.

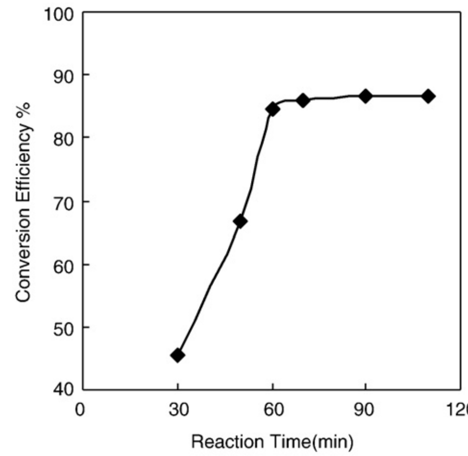

Fig. 4 – Effect of reaction time on WCO conversion efficiency. Under 6:1 methanol/oil molar ratio, 1.0 wt.% NaOH, at 50 °C.

3.4. *Effect of reaction temperature*

The effect of reaction temperature on WCO conversion efficiency is shown in Fig. 5. Transesterification can occur at different temperatures depending on the oil used. The maximum WCO conversion efficiency was obtained at 50 °C temperature, and other researchers [17,18] achieved better results at temperatures above 50 °C and up to 70 °C while using refined linseed oil and brassica carinata oil, respectively. Results revealed that the WCO conversion efficiency decreased when other reaction temperatures were used. When the reaction temperature closes or exceeds the boiling point of methanol, the methanols will vaporize and form a large number of bubbles then inhibit the reaction. Although a reflux condenser was used in the experiment to avoid methanol losses, the WCO conversion efficiency significantly decreased at temperatures more than 60 °C. As well as

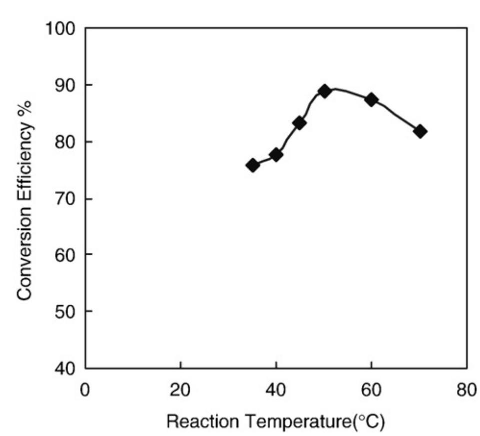

Fig. 5 – Effect of temperature on WCO conversion efficiency. Under 6:1 methanol/oil molar ratio 1.0 wt.% NaOH, reacted 60 min.

Table 3 – The experimental analyses of orthogonal test

NO.	A methanol/oil molar ratio	B NaOH amount (wt.%)	C Reaction temperature (°C)	D Reaction time (min)	Conversion (%)
1	3:1	0.7	35	30	39.1
2	3:1	1.0	50	60	76.9
3	3:1	1.3	60	90	45.9
4	6:1	0.7	50	90	85.5
5	6:1	1.0	60	30	66.8
6	6:1	1.3	35	60	44.70
7	9:1	0.7	60	60	40.70
8	9:1	1.0	35	90	72.10
9	9:1	1.3	50	30	85.8
$Kmean_{1j}$	53.967	55.100	51.967	63.900	
$Kmean_{2j}$	65.667	71.933	82.733	54.100	
$Kmean_{3j}$	66.200	58.800	51.133	67.833	
Range	12.233	16.833	31.600	13.733	
The optimum	A_3	B_2	C_2	D_3	

* $Kmean_{ij}$: indicate that the average value of *j* (conversion) on *i* factor.

economic reasons considered, 50 °C was selected during transesterification.

3.5. *Orthogonal test*

Extensive preliminary experiments with WCO samples indicated that the major variables in the transesterification process are: methanol/oil molar ratio, catalyst amount, reaction time and reaction temperature. Based on the result of the preliminary experiments, a four-factor (A methanol/oil molar ratio, B NaOH amount, C Reaction temperature and D Reaction time) and three-level (for A 3:1, 6:1, 9:1; for B 0.7 wt.%, 1.0 wt.%, 1.3 wt.%; for C 35 °C, 50 °C, 60 °C and for D 30 min, 60 min, 90 min) orthogonal test was designed to determine optimum conditions for the transesterification process of WCO. The analyzed results of orthogonal test about the main effect of the four factors are shown in Table 3. The analyzed results show that the reaction temperature is the most important factor, followed by catalyst amount, reaction time and the methanol/oil molar ratio. The optimum experimental conditions, which were obtained from the orthogonal test, were methanol/oil molar ratio 9:1, with 1.0 wt.% sodium hydroxide, temperature of 50 °C and 90 min. Afterwards, some verified experiments were done to testify the results. It turned out that when methanol/oil molar ratio 6:1, at1.0 wt.% sodium hydroxide, 50 °C and reacted 90 min, the satisfied results can be achieved and the WCO conversion efficiency led to 89.8% which showed methanol/oil molar ratio 6:1 is more suitable in the process.

Table 4 – The properties of biodiesel sample compared to diesel fuel and EN14214 biodiesel standard

Parameter	Samples	Diesel fuel	EN14214
Density (15 °C, kg/m^3)	890	NA	860–900
Flash point (°C)	171	>65	>101
Kinematic viscosity (40 °C, mm^2/s)	4.23	3.0–8.0	3.5–5.0
Sulfur content (wt.%)	0.007	<0.05	<0.01
10% Conradson carbon residue	0.2	0.3	0.3
Copper strip corrosion (3 h, 50 °C)	1a	class1	class1
Water content (mg/Kg)	150	NA	<500
Cold filter plugging point (°C)	1	≤4	NA
Free glycerol (%)	0.008	NA	0.02
Total glycerol (%)	0.21	NA	0.25
Acid value (mg KOH/g)	0.48	<0.1	≤0.5
Cetane number	54.5	>49	≥51
Caloric value (MJ/kg)	32.9	41.8	NA

NA stands for not available.

3.6. *The properties of biodiesel*

Standards and quality control of manufacturing and distribution of biodiesel are being developed to assure that reliable and consistent fuels are supplied to users. As an alternative fuel, biodiesel has physical and chemical properties qualifying to the operation of diesel engines. These properties play a vital role in quality control in the petroleum-based diesel fuel industry.

Biodiesel derived from WCO was analyzed by Petroleum products Analysis Institute of Tianjin City and the results are shown in Table 4. As is indicated the biodiesel sample meet EN14214 standards for density, kinematic viscosity, copper corrosion, acid value, cetane number, free glycerol and total

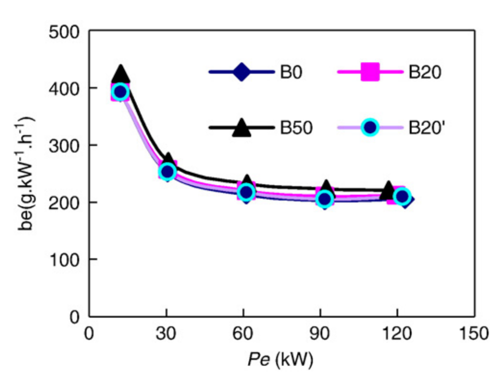

Fig. 6 – Fuel consumption per hour at 1300 r/min.

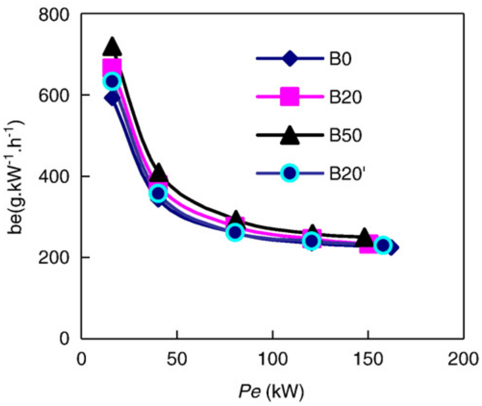

Fig. 7 – Fuel consumption per hour at 2200 r/min.

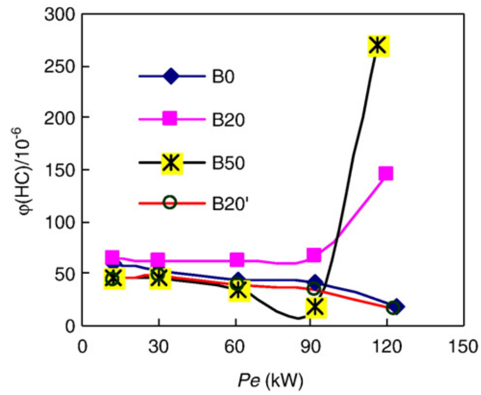

Fig. 9 – HC emissions at 1300 r/min.

glycerol. There was slight difference in density and viscosity comparing with 0# diesel but completely acceptable. The higher flash-point (171 °C) of biodiesel sample is beneficial in safety aspect, and the low sulfur content (0.007 mg/kg) is the reason for the extremely low SOx emission associated with its use as fuel. The Cetane number (54.5) is higher than 0# diesel resulting in a smoother running of the engine with less noise. Biodiesel sample is an oxygenated fuel naturally with oxygen content about 10% which contributes to the favorable emission, but leads to a little bit low caloric value (32.9 MJ/kg) compared with petro-diesel. Biodiesel nearly meets all the properties of normal diesel fuel, according to 0# diesel and EN 14214 standards, which indicates that Biodiesel derived from WCO has adequate values compared to diesel fuel.

3.7. *Engine performance*

3.7.1. *Fuel consumption*

The fuel consumption under two typical brake specific 1300 r/min and 2200 r/min are shown in Figs. 6 and 7, which indicated for all fuels tested, brake specific fuel consumption decreased with increasing in load. One possible explanation for this reduction could be due to the higher percentage of increase in brake power with load as compared to fuel consumption. In general, the reference fuel B0 resulted in smaller fuel consumption per unit energy output compared to all other blends fuels which due to the lower calorific value in the blend fuels. However, considering in the other way, such as at brake specific of 1300 r/min, minimal fuel consumption for B0, B20′, B20 and B50 was 203.4 g/(kW·h), 205.4 g/(kW·h), 210.7 g/(W·h) and 222.8 g/(kW·h) respectively, with most economical power 122.9 kW, 121.9 kW, 119.5 kW and 116.6 kW, which showed that blend fuels have little differences in engine performance.

3.7.2. *Exhaust emissions*

The emissions of carbon monoxide, unburned hydrocarbons and nitrogen oxide were examined and the results are shown in Figs. 8–10 for all the test fuels. B20, B50 blend fuels both were inferior to the reference fuel B0 as far as carbon monoxide and unburned hydrocarbons were concerned and better in nitrogen oxide emission. This general trend may be partially due to the presence of methanol in crude biodiesel which result in low cetane number value. Because carbon monoxide and unburned hydrocarbons are the products of incomplete combustion, the lower cetane number of blend fuels results in lower tendency to form ignitable mixture, and thus, higher carbon monoxide and unburned hydrocarbons [19,20]. However, refined biodiesel blend fuel B′20 consistently gave the minimum amounts of all emissions considered. CO and HC were reduced by 18.6% and 26.7%, respectively. The total particles emission for B′20 was 0.0714 g/(kW·h) and was reduced by 20.58% compared to the reference fuel B0 which had 0.0899 g/(kW·h) total particles emission.

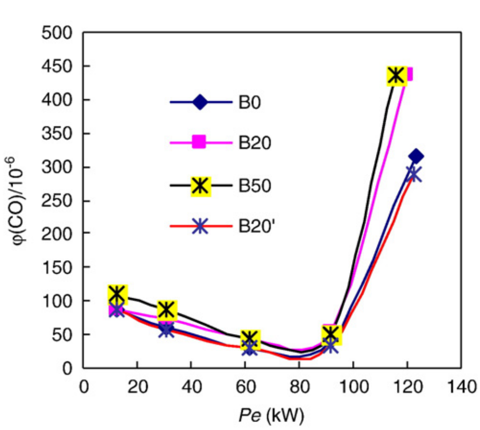

Fig. 8 – CO emissions at 1300 r/min.

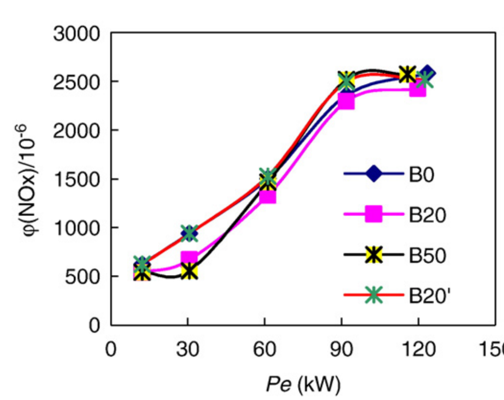

Fig. 10 – NOx emissions at 1300 r/min.

4. Conclusions

The WCO after pretreatment can have a transesterification reaction. Compared to pure oils, the conversation rate can be reduced. With the appropriate quality upgrading treatment, biodiesel obtained from WCO can be used as a fuel in diesel engines. Through the feasible analysis of orthogonal test, biodiesel of good quality can be produced from WCO in the following reaction conditions: methanol/oil 9:1 molar ratio, with1.0 wt.% sodium hydroxide, temperature at 50 °C and 90 min. Verified experiments showed methanol/oil molar ratio 6:1 was more suitable in the process. Engine testing results showed without any modification to diesel engine, under all conditions dynamical performance kept normal, and the B20, B50 blend fuels led to unsatisfactory emissions whilst the B′20 blend fuels reduced significantly particles, HC and CO emissions. For example CO, HC and particles were reduced by 18.6%, 26.7% and 20.58%. China has so many kinds of waste oils for biodiesel production. And the government should make full use of these resources and ensure national diesel security as a long lasting supplement.

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